

2nd Order Homogeneous ODEs

Say we have the below 2nd order homogeneous ODE:

$$y'' + py' + qy = 0$$

where p, q are real constants.

The characteristic equation of the above ODE is:

$$\lambda^2 + p\lambda + q = 0$$

which has the roots:

$$\lambda_{1,2} = -\frac{p}{2} \pm \sqrt{\Delta} \quad \text{where } \Delta = \frac{p^2 - 4q}{4}$$

Now there are 3 cases:

1. If $\Delta > 0$, the general solution is

$$y(x) = (c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x})$$

where c_1, c_2 are real constant ($\lambda_1, \lambda_2 \in R$) and $\lambda_1 \neq \lambda_2$.

2. If $\Delta = 0$, we have $\lambda_1 = \lambda_2 = -\frac{p}{2}$. Then the general solution is

$$y(x) = (c_1 + c_2 x) e^{-\frac{px}{2}}$$

3. $\Delta < 0$, we have the complex roots $\lambda_{1,2} = -\frac{p}{2} \pm i\sqrt{-\Delta} = \alpha \pm i\beta$ where $\sqrt{-\Delta} = \frac{4q-p^2}{4} > 0$.

Then the general solution is

$$y(x) = e^{-\frac{px}{2}} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

Non-Homogeneous

$$y'' + p(x)y' + q(x)y = f(x)$$

The general solution for this ODE is

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y^*(x)$$

where

$$y^*(x) = \int_{x_0}^x K(x, t) f(t) dt$$

with

$$K(x, t) = \frac{\begin{vmatrix} y_1(t) & y_1(x) \\ y_2(t) & y_2(x) \end{vmatrix}}{\begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix}}$$

Uniqueness

Let's say you have the ODE:

$$\begin{cases} y'' + p(x)y' + q(x)y = f(x) \\ y(a) = y(b) = 0 \end{cases}$$

where $p(x), q(x), f(x) \in C[a, b]$. Now, if $f(x) = 0$, we want to derive the solution to the above equation and proof its uniqueness.

For that, let's assume there are two linearly unrelated (线性无关) solutions $\phi(x)$ and $\psi(x)$ to the homogeneous equation:

$$y'' + p(x)y' + q(x)y = 0$$

so then the general solution to that equation would be

$$y(x) = C_1\phi(x) + C_2\psi(x) + y^*(x)$$

where C_1, C_2 are any constants.

$y^*(x)$ is defined as:

$$y^*(x) = \int_{x_0}^x G(x, t)f(t)dt$$

where

$$G(x, t) = \frac{\begin{vmatrix} \phi(t) & \phi(x) \\ \psi(t) & \psi(x) \end{vmatrix}}{\begin{vmatrix} \phi(t) & \psi(t) \\ \phi'(t) & \psi'(t) \end{vmatrix}}$$

Because $y(x)$ satisfies $y(a) = y(b) = 0$, we can write the expression:

$$\begin{cases} C_1\phi(a) + C_2\psi(a) = -y^*(a) \\ C_1\phi(b) + C_2\psi(b) = -y^*(b) \end{cases}$$

The determinant of the above relation is

$$\Delta = \begin{vmatrix} \phi(a) & \psi(a) \\ \phi(b) & \psi(b) \end{vmatrix}$$

And then we can prove that $\Delta \neq 0$:

- Otherwise, there would be a constant c such that $\psi(a) = c\phi(a)$ and $\psi(b) = c\phi(b)$
- If $c = 0$, then $\psi(a) = \psi(b) = 0$
 - This would mean $\psi(x) \equiv 0$, but that would contradict the assumption that $\psi(x)$ is a non-zero solution of the ODE
- For the case $c \neq 0$, let $z(x) = \phi(x) - \frac{\psi(x)}{c}$.
 - It is easy to prove that $z(a) = z(b) = 0$
 - Let $z(x)$ be a solution to the ODE, then from the assumption we know $z(x) \equiv 0$, so $\psi(x) = c\phi(x)$
 - But $\psi(x) = c\phi(x)$ is contradicting the linearly independent assumption
- Therefore, $\Delta \neq 0$.

Then, we can say that the only solution to C_1 and C_2 is:

$$\begin{cases} C_1 = \frac{y^*(b)\psi(a) - y^*(a)\psi(b)}{\Delta} \\ C_2 = \frac{y^*(a)\phi(b) - y^*(b)\phi(a)}{\Delta} \end{cases}$$